- Paper 1 Non-calculator, Paper 2 Calculator
- Grades C-G available
- Available in June and November
- 30-40% of each paper assesses the Functional elements of mathematics

References

Each topic in this qualification contains a specification reference (for example, **SP a** for Statement a, Statistics and Probability), the content descriptor and examples of concepts and skills associated with that content descriptor.

Foundation

1 Number

What students need to learn:

Ref	Content descriptor	Concepts and skills
N a	Add, subtract, multiply and divide any number	• Add, subtract, multiply and divide whole numbers, integers, negative numbers, decimals, fractions and numbers in index form • Recall all multiplication facts to 10×10, and use them to derive quickly the corresponding division facts • Multiply or divide any number by powers of 10 • Multiply or divide by any number between 0 and 1 • Solve a problem involving division by a decimal (up to two decimal places) • Write numbers in words • Write numbers from words • Recall the fraction-to-decimal conversion of familiar fractions
N b	Order rational numbers	• Order integers, decimals and fractions • Understand and use positive numbers and negative integers, both as positions and translations on a number line
N c	Use the concepts and vocabulary of factor (divisor), multiple, common factor, Highest Common Factor (HCF), Least Common Multiple (LCM), prime number and prime factor decomposition	• Recognise even and odd numbers • Identify factors, multiples and prime numbers • Find the prime factor decomposition of positive integers • Find the common factors and common multiples of two numbers • Find the Lowest common multiple (LCM) and Highest common factor (HCF) of two numbers

Ref	Content descriptor	Concepts and skills
N d	Use the terms square, positive and negative square root, cube and cube root	• Recall integer squares up to 15 × 15 and the corresponding square roots • Recall the cubes of 2, 3, 4, 5 and 10 • Find squares and cubes • Find square roots and cube roots
N e	Use index notation for squares, cubes and powers of 10	• Use index notation for squares and cubes • Use index notation for powers of 10 • Find the value of calculations using indices
N f	Use index laws for multiplication and division of integer powers	• Use index laws to simplify and calculate the value of numerical expressions involving multiplication and division of integer powers, and of powers of a power (NB: Fractional, zero and negative powers are only assessed on Higher Tier)
N h	Understand equivalent fractions, simplifying a fraction by cancelling all common factors	• Find equivalent fractions • Write a fraction in its simplest form • Convert between mixed numbers and improper fractions • Compare fractions
N i	Add and subtract fractions	• Add and subtract fractions
N j	Use decimal notation and recognise that each terminating decimal is a fraction	• Understand place value • Identify the value of digits in a decimal or whole number • Write terminating decimals as fractions • Recall the fraction-to-decimal conversion of familiar simple fractions • Convert between fractions and decimals
N k	Recognise that recurring decimals are exact fractions, and that some exact fractions are recurring decimals	• Recognise that recurring decimals are exact fractions, and that some exact fractions are recurring decimals • Convert between recurring fractions and decimals

Foundation

Ref	Content descriptor	Concepts and skills
N l	Understand that 'percentage' means 'number of parts per 100' and use this to compare proportions	• Order fractions, decimals and percentages • Convert between fractions, decimals and percentages
N m	Use percentage	• Use percentages to solve problems • Find a percentage of a quantity in order to increase or decrease • Use percentages in real-life situations – VAT – Value of profit or loss – Simple Interest – Income tax calculations (NB: Repeated proportional change is only assessed at Higher Tier)
N o	Interpret fractions, decimals and percentages as operators	• Find a fraction of a quantity • Express a given number as a fraction of another • Find a percentage of a quantity • Express a given number as a percentage of another number • Use decimals to find quantities • Understand the multiplicative nature of percentages as operators • Use a multiplier to increase or decrease by a percentage in any scenario where percentages are used
N p	Use ratio notation, including reduction to its simplest form and its various links to fraction notation	• Use ratios • Write ratios in their simplest form

Ref	Content descriptor	Concepts and skills
N q	Understand and use number operations and the relationships between them, including inverse operations and hierarchy of operations	• Multiply and divide numbers using the commutative, associative, and distributive laws and factorisation where possible, or place value adjustments • Use inverse operations • Use brackets and the hierarchy of operations • Use one calculation to find the answer to another • Understand 'reciprocal' as multiplicative inverse, knowing that any non-zero number multiplied by its reciprocal is 1 (and that zero has no reciprocal because division by zero is not defined) • Find reciprocals • Understand and use unit fractions as multiplicative inverses • Solve word problems
N t	Divide a quantity in a given ratio	• Divide a quantity in a given ratio • Solve a ratio problem in context
N u	Approximate to specified or appropriate degrees of accuracy including a given power of ten, number of decimal places and significant figures	• Round numbers to a given power of 10 • Round to the nearest integer and to any given number of significant figures • Round to a given number of decimal places • Estimate answers to calculations, including use of rounding
N v	Use calculators effectively and efficiently, including statistical functions	• Know how to enter complex calculations • Enter a range of calculations including those involving time and money • Understand and interpret the calculator display, knowing when the display has been rounded by the calculator, and know not to round during the intermediate steps of a calculation • Use a range of calculator functions including $+$, $-$, $\times$, $\div$, x^2, $\sqrt{x}$, memory, x^y, $x^{1/y}$ and brackets